

Noise Analysis — 10um

PAGE 1 — FILE STATISTICS

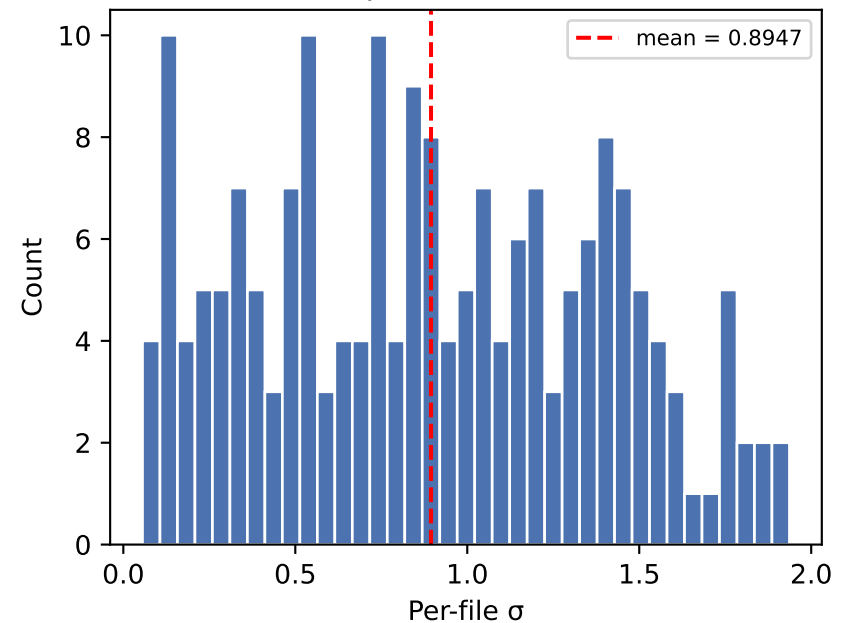
Folder: data/dataset_hamming_c1/test/10um
Files: 189
Signal length: 16384–16384 samples
Dtype: float64

Aggregate noise statistics:

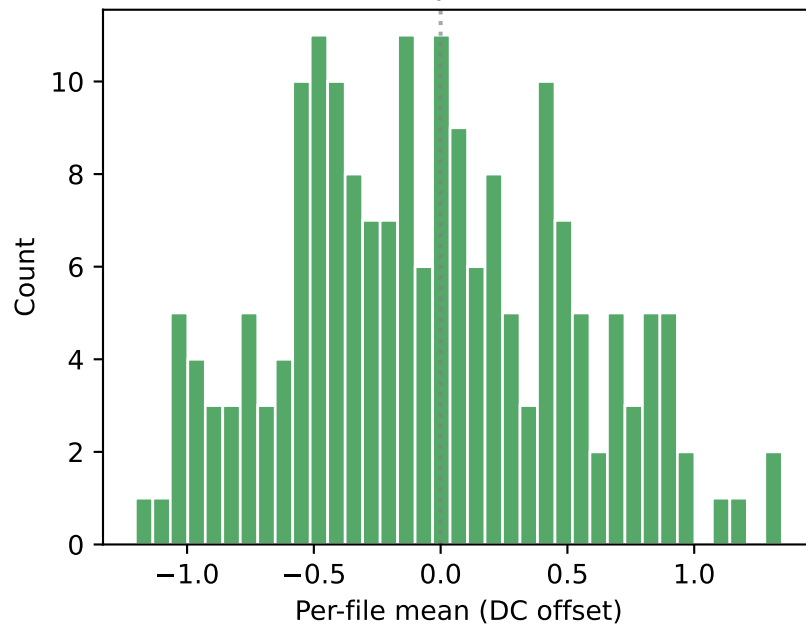
mean(σ): 0.894657
std(σ): 0.491730
CV(σ): 0.5496
mean(RMS): 1.063060
mean(kurtosis): -0.4236 ($\emptyset$ = Gaussian)

mean(DC): -0.048772
std(DC): 0.549892
Amplitude range: [-2.4065, 5.2752]

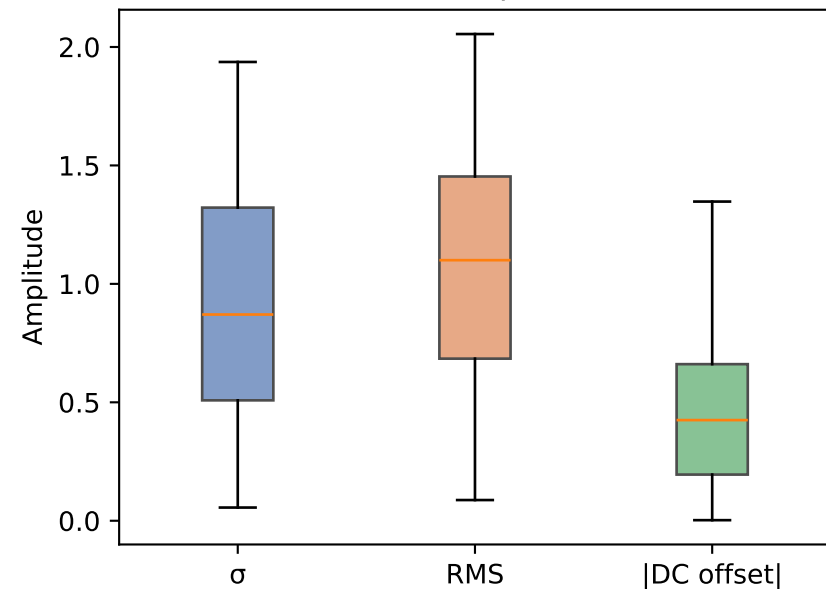
Distribution of per-file standard deviations



Distribution of per-file DC offsets

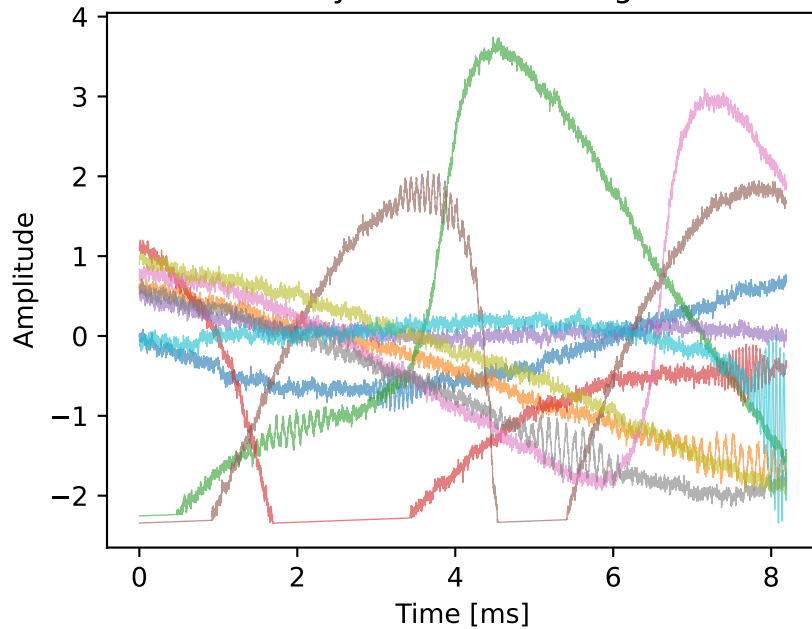


Distribution of amplitude metrics

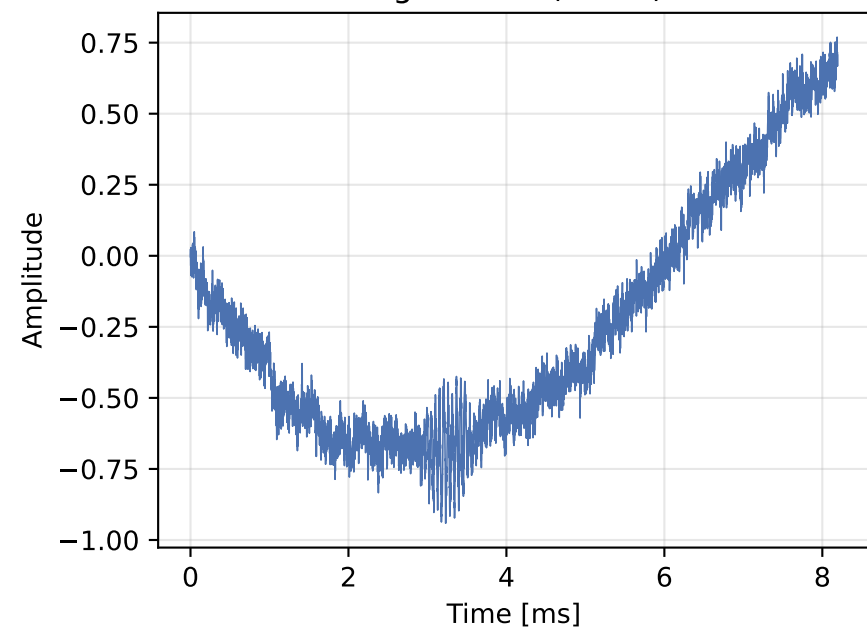


Time-Domain Analysis — 10um

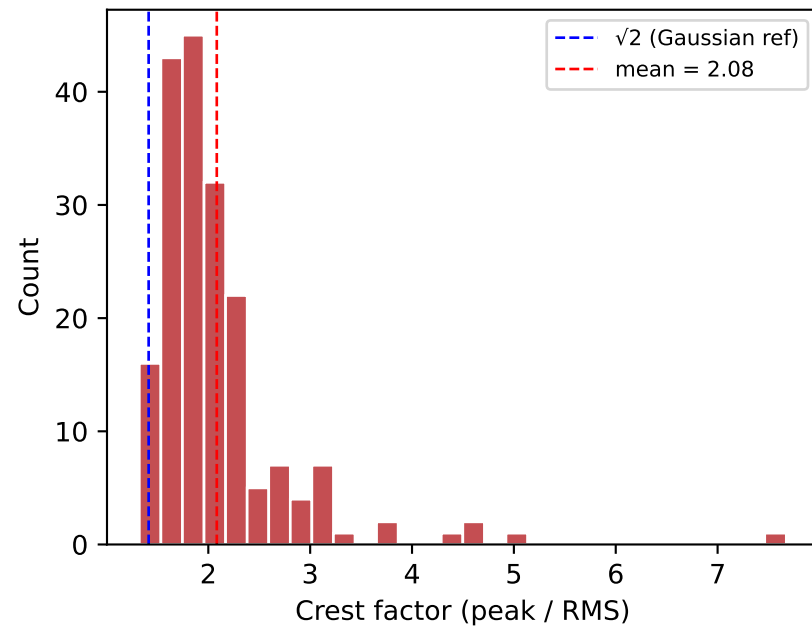
Overlay of 10 random signals



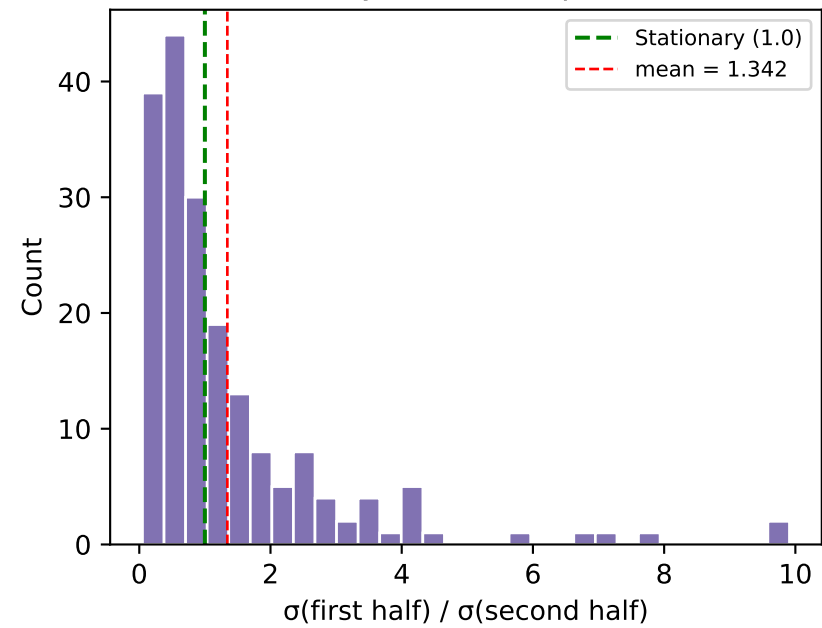
Signal #15 (detail)



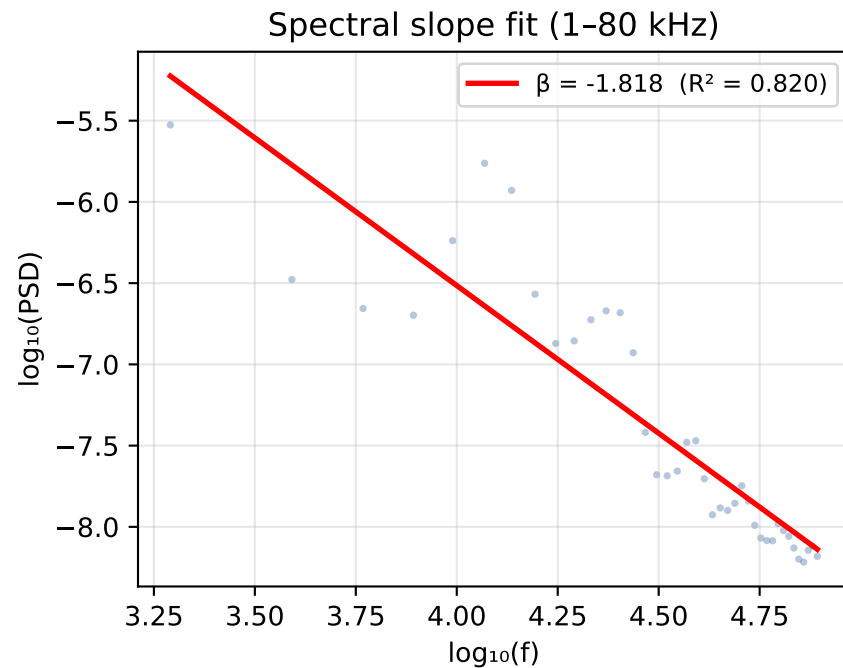
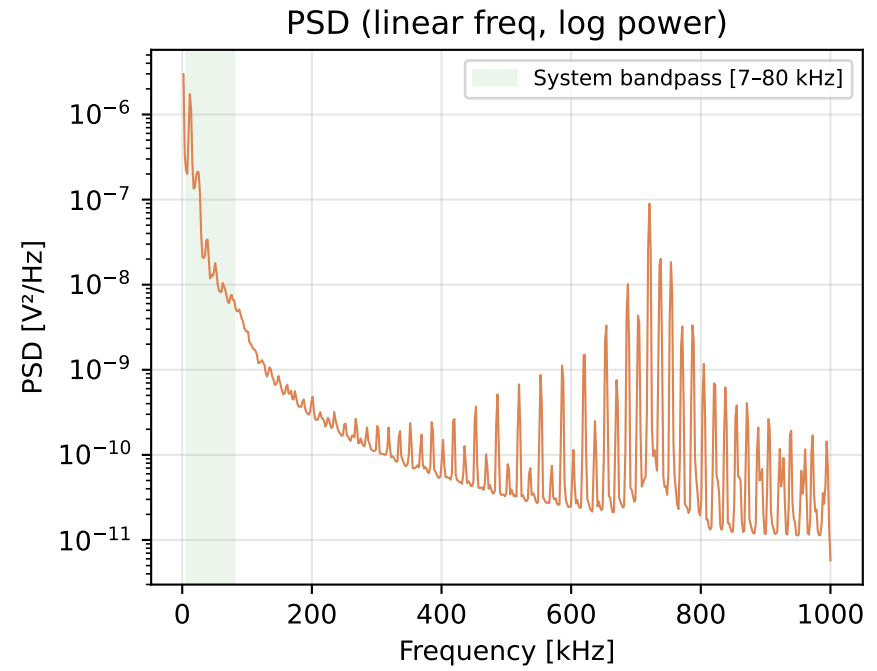
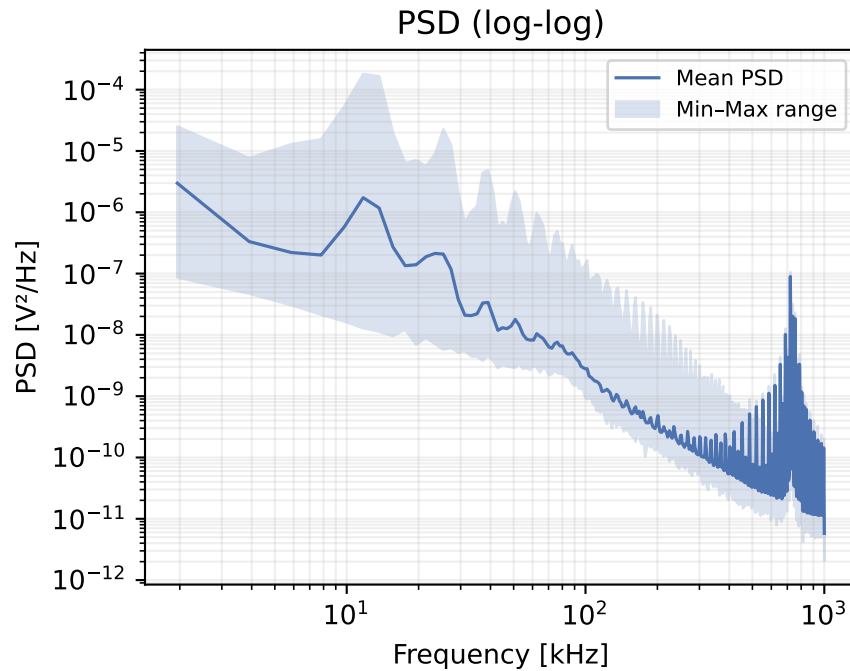
Crest factor distribution



Stationarity test (half-split σ ratio)



Power Spectral Density — 10um



SPECTRAL ANALYSIS SUMMARY

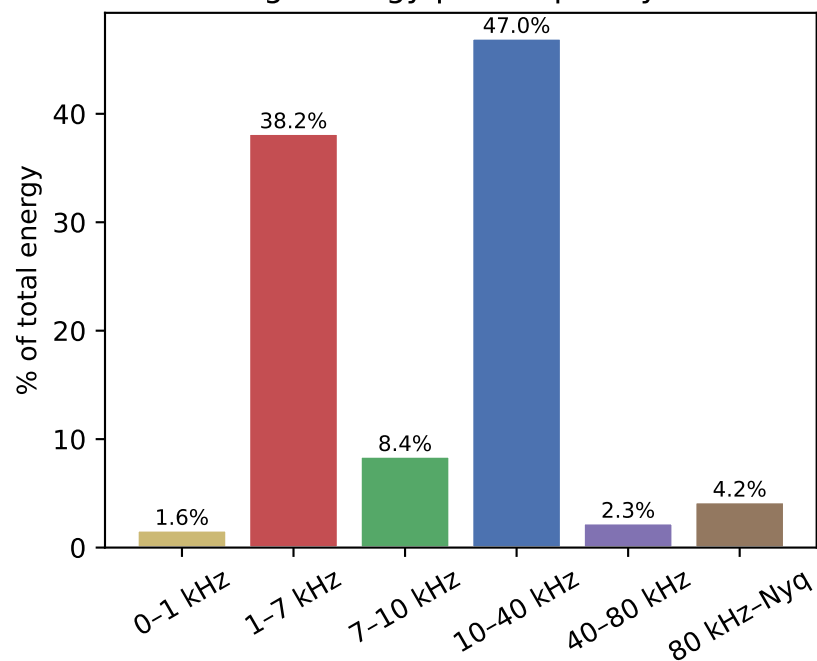
Spectral slope β : -1.8181
Fit quality R^2 : 0.8200
Spectral flatness: 0.0098 (1.0 = white)

Slope interpretation:
 $\beta \approx 0 \rightarrow$ White noise (flat PSD)
 $\beta \approx -1 \rightarrow$ Pink / $1/f$ (equal energy per octave)
 $\beta \approx -2 \rightarrow$ Brown / $1/f^2$ (random walk)

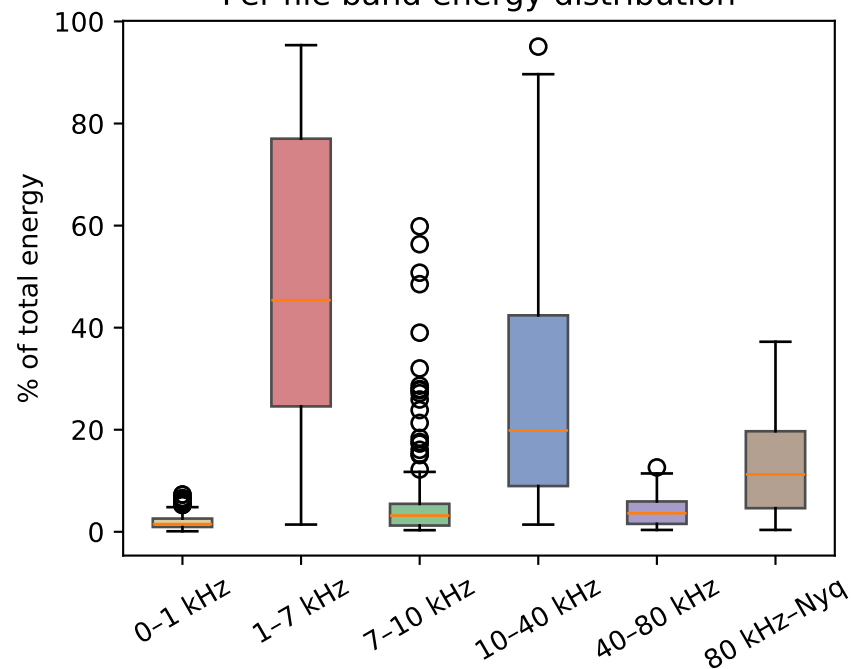
Classification: Brown ($1/f^2$)

Frequency Band Energy — 10um

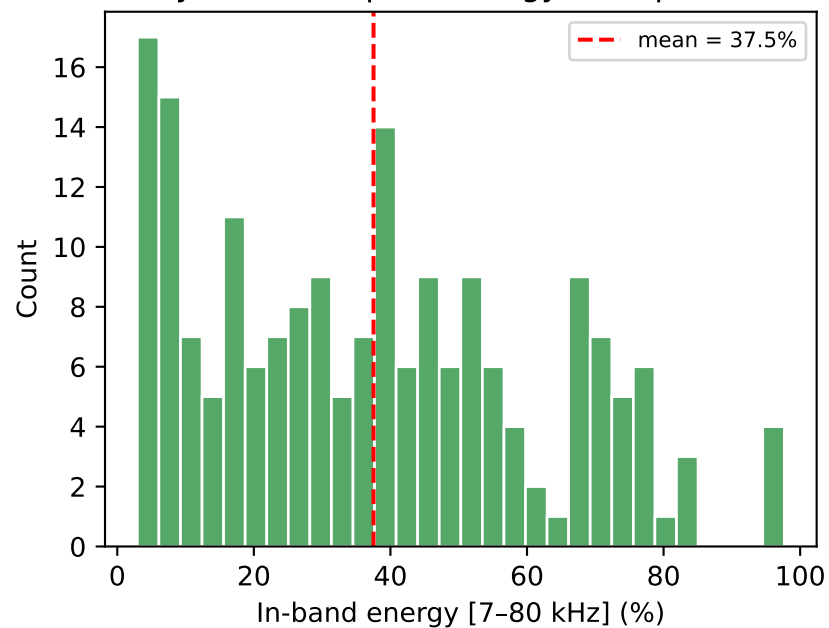
Average energy per frequency band



Per-file band energy distribution



System bandpass energy ratio per file



BAND ENERGY SUMMARY

Band	Mean %	Std %
0-1 kHz	1.6%	1.5%
1-7 kHz	38.2%	28.2%
7-10 kHz	8.4%	9.5%
10-40 kHz	47.0%	21.9%
40-80 kHz	2.3%	3.1%
80 kHz-Nyq	4.2%	10.0%
In-band [7-80k]	37.5%	24.4%

Noise Type Classification — 10um

NOISE TYPE CLASSIFICATION

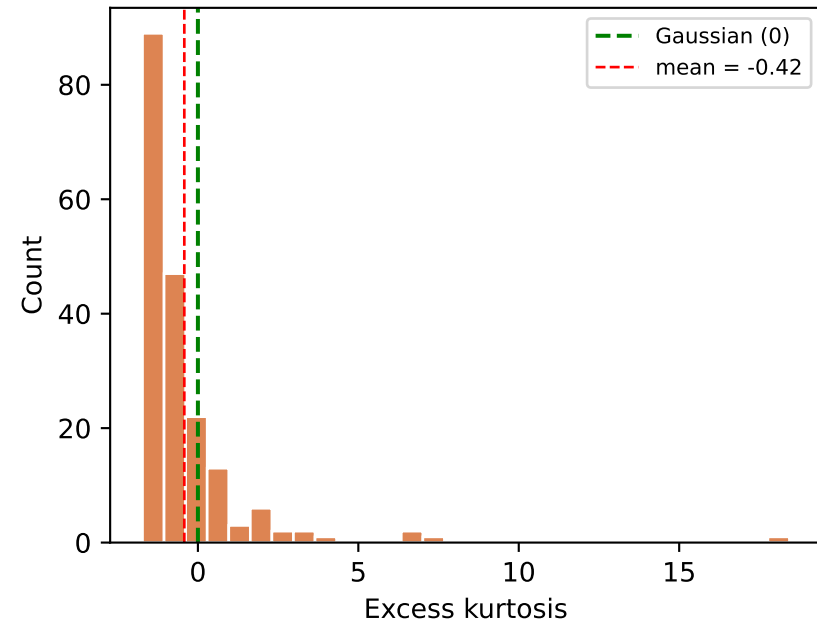
Spectral color: Brown ($1/f^2$)
slope β = -1.8181
flatness = 0.0098

Amplitude dist.: Gaussian
kurtosis (mean): -0.4236
(excess kurtosis: 0 = Gaussian,
>0 = heavy-tailed, <0 = light-tailed)

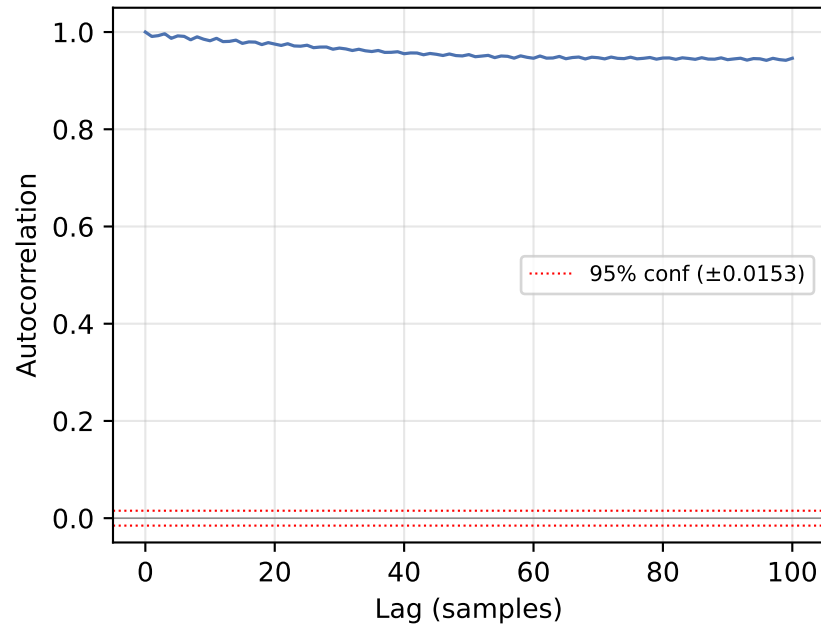
Decision boundaries:
 $|\beta| < 0.3$ → White
0.3–1.5 → Pink ($1/f$)
> 1.5 → Brown ($1/f^2$)

 $|\text{kurt}| < 0.5$ → Gaussian
0.5–2.0 → Near-Gaussian
> 2.0 → Non-Gaussian

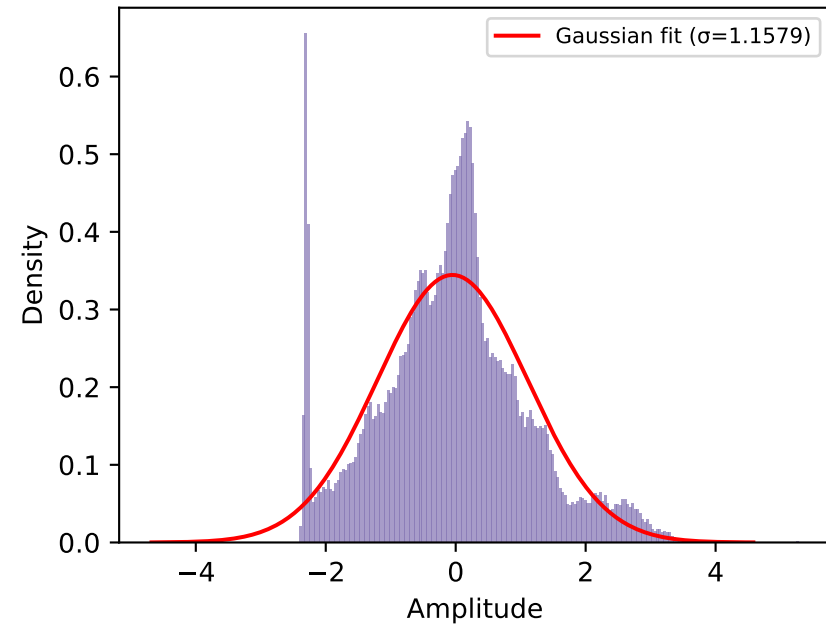
Per-file kurtosis distribution



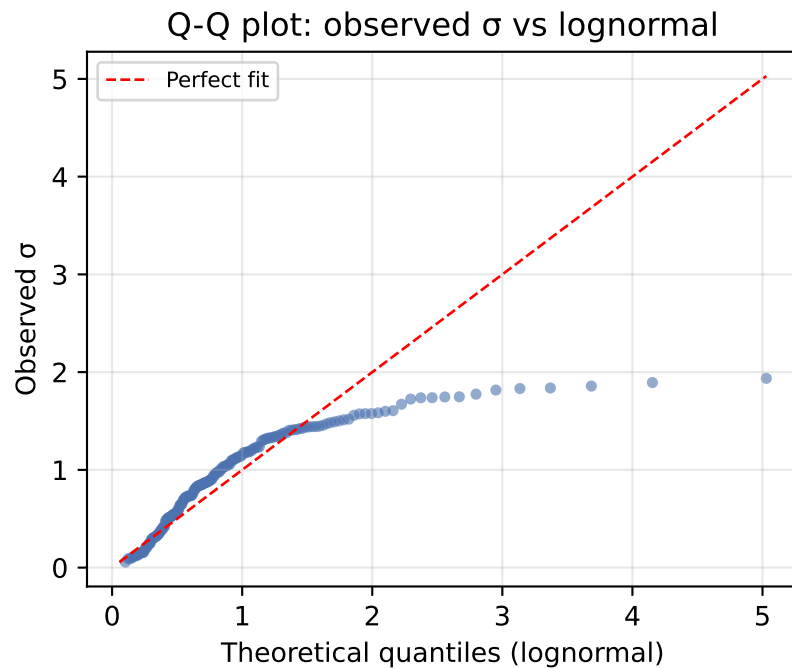
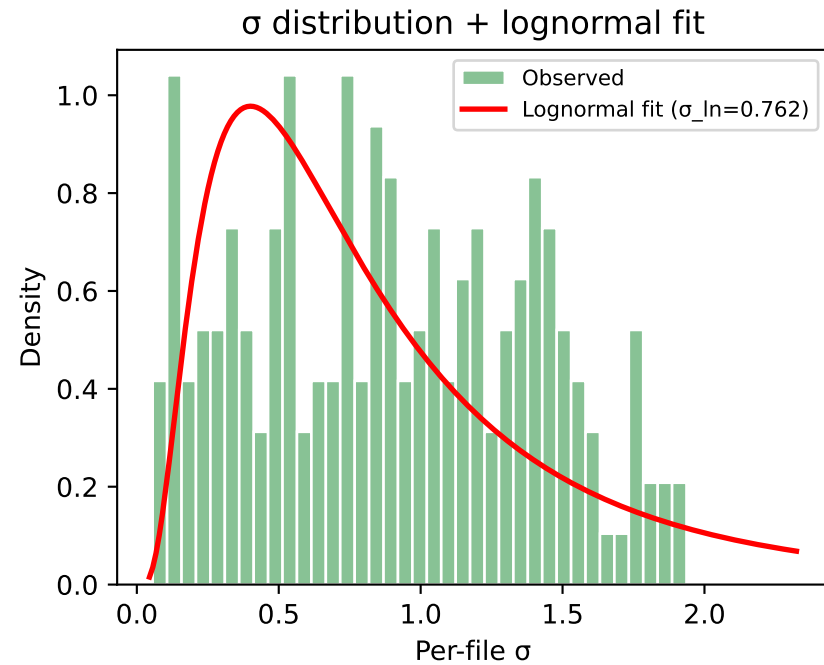
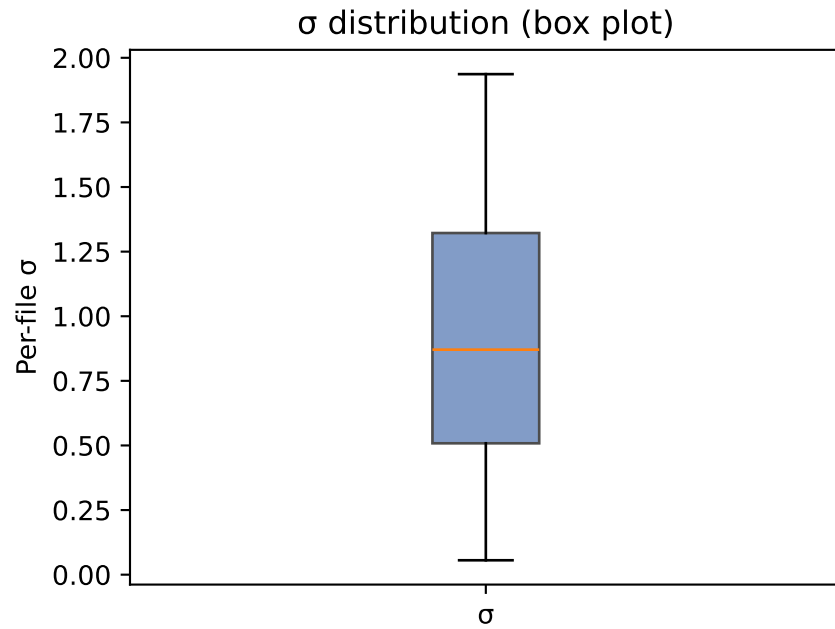
Average autocorrelation (first 100 lags)



Amplitude distribution (all samples pooled)



Amplitude Variability (Inter-file) — 10um



AMPLITUDE VARIABILITY SUMMARY

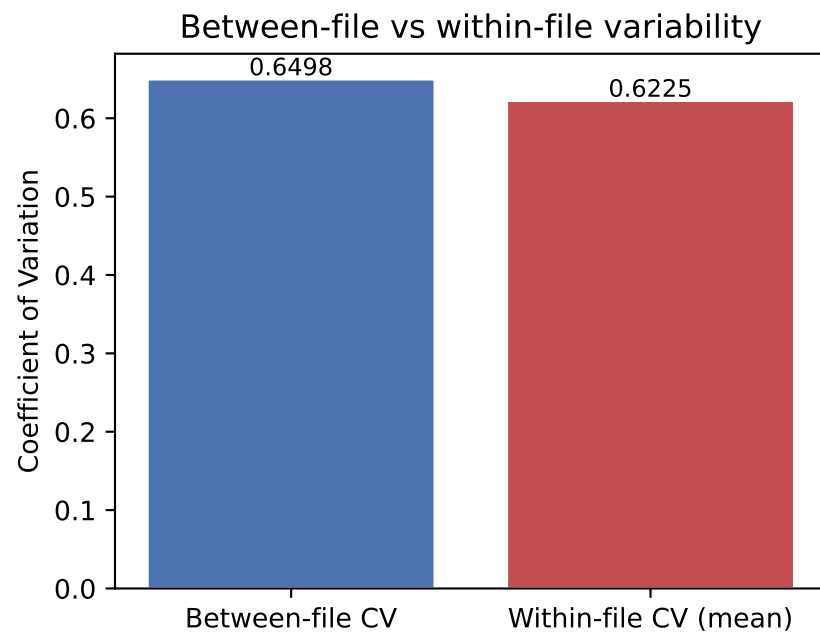
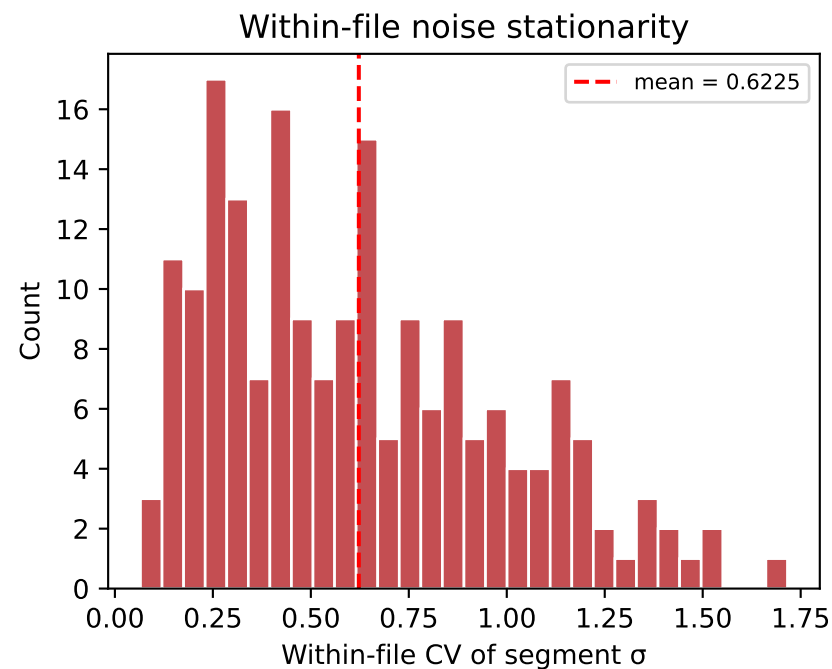
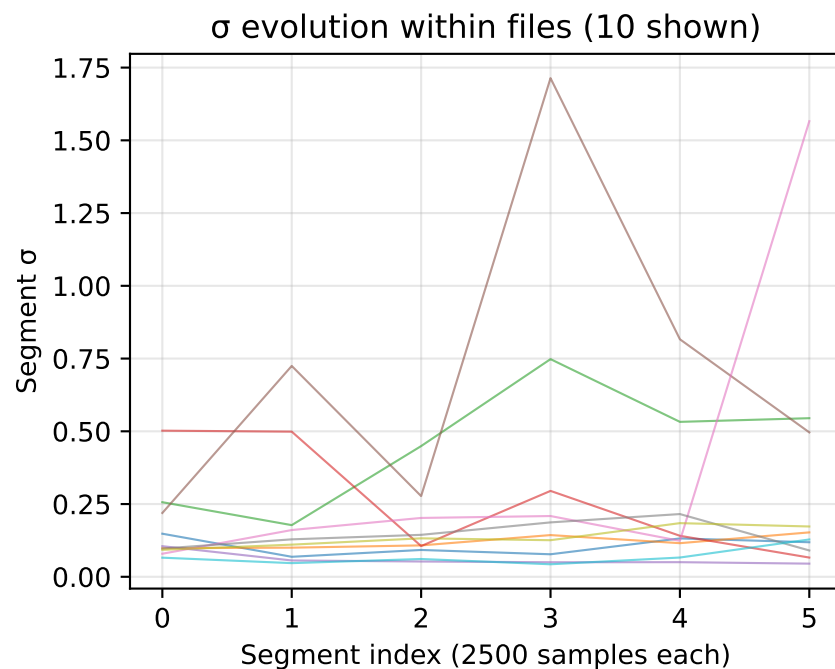
Number of files: 189
mean(σ): 0.894657
std(σ): 0.491730
CV = std(σ)/mean(σ): 0.5496

Lognormal fit parameters:
shape ($\sigma_{\ln}$): 0.7621
scale ($\exp(\mu_{\ln})$): 0.715907

Reference (from real noise files):
CV_{ref} = 0.19
Ratio CV/CV_{ref}: 2.89

Interpretation:
High variability – mixed conditions or signal content

Segment-Level Analysis — 10um



SEGMENT ANALYSIS SUMMARY

Segment length: 2500 samples
Files analyzed: 189 / 189
Segments per file: 6–6

Between-file CV: 0.6498
Within-file CV (mean): 0.6225
Within-file CV (std): 0.3622

Interpretation:
Significant non-stationarity – noise level
varies within individual recordings
Within-file variability is comparable to
between-file variability